Does ETF Momentum Survive an Honest Out-of-Sample Test?

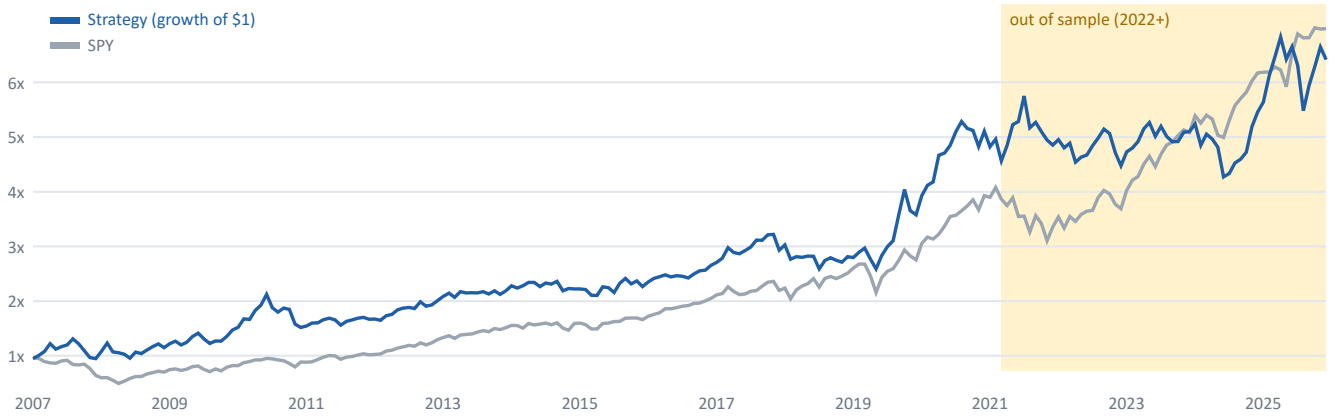
Denver Lo | October 16, 2026 | Quant research note | Code: quant_study.py (Python: pandas, NumPy, SciPy) | Data: 2007-11-30 to 2026-10-16, 17 ETFs, monthly

Answer: no, not once the parameters are chosen without peeking. Picking the best of 15 configurations on 2007-2021 data alone selects a 126-day lookback holding the top 3. Out of sample (2022 to October 2026) it earned a Sharpe of **0.17** against SPY's 0.55, with a Newey-West alpha of -1.6% a year (t = -0.19). Across the grid, in-sample rank predicted out-of-sample rank *negatively* (Spearman rho = -0.44). The 252-day/top-2 version that topped my own Project Hero One tournament (Sharpe 0.90 there) scores 0.70 here, and it was the winner only because the tournament ranked candidates on held-out data. That is selection bias.

Strategy and test design

- **Rule (defensive dual momentum):** at each month-end, rank 15 risk ETFs (equity, sector, credit, REIT, commodity) by trailing return. Hold the top *k* equally weighted, but only those beating T-bills over the same window (absolute momentum); empty slots earn the T-bill rate. If none qualify, hold the stronger of TLT and GLD.
- **No look-ahead:** the signal uses the month-end close and is applied to the following month. Universe: SPY, QQQ, IWM, DIA, TLT, GLD, SLV, HYG, VNQ, XLE, XLF, XLK, XLV, XLI, XLB, DBA, USO (dividend-adjusted). Cash and Sharpe use the FRED 3-month T-bill.
- **Costs:** 5bp per side on actual turnover (8.6x a year), stressed to 10bp.
- **Overfitting controls:** a 15-point grid (lookback 63-315 days x top 1-3) searched on the training window only; deflated Sharpe ratio for multiple testing; probabilistic Sharpe ratio; Newey-West (3-lag) alpha t-stats; a 5,000-draw circular block bootstrap (6-month blocks) of excess returns over SPY.

Results



	Strategy, train	SPY, train	Strategy, test	Strategy, test, 2x costs	SPY, test	60/40, test
CAGR	12.0%	10.4%	5.5%	5.0%	11.8%	3.4%
Volatility	18.7%	15.6%	18.3%	18.3%	15.6%	13.6%
Sharpe (excess of T-bills)	0.67	0.68	0.17	0.15	0.55	0.02
Sortino	1.02	0.87	0.26	0.22	0.87	0.04
Max drawdown	-28.6%	-48.8%	-25.7%	-26.7%	-20.2%	-22.5%
Calmar	0.42	0.21	0.21	0.19	0.58	0.15
Positive months	60.6%	66.5%	63.8%	63.8%	63.8%	60.3%

Statistical tests	Train	Test
Alpha vs SPY, annualized (Newey-West t)	5.9% (1.39)	-1.6% (-0.19)
Beta to SPY	0.62	0.56
Deflated Sharpe: bar for best of 15 trials / P(skill)	0.24 / 94.0%	

Lookback \ top-k	1	2	3
63 days	0.16 / 0.42	0.20 / 0.53	0.42 / 0.35
126 days	0.36 / 0.39	0.53 / 0.27	**0.67 / 0.17**
189 days	0.26 / 0.76	0.31 / 0.67	0.39 / 0.54
252 days	0.26 / 0.97	0.41 / 0.70	0.41 / 0.78

Probabilistic Sharpe: P(SR > 0)	64.1%
Probabilistic Sharpe: P(SR > SPY's)	20.9%
Bootstrap P(mean excess over SPY > 0)	27.7%
Grid configs beating SPY's test Sharpe	40.0%

315 days	0.14 / 0.63	0.33 / 0.47	0.38 / 0.30

Sharpe, train / test, for each configuration. Highlighted: the configuration chosen on training data. Longer lookbacks did better after 2021 but worse before it. No rule known in 2021 would have pointed to them (rho = -0.44, p = 0.10).

Reading it. In training, the strategy matched SPY's Sharpe (0.67 vs 0.68) with a much smaller drawdown (-28.6% vs -48.8%): it sidestepped 2008. Its deflated-Sharpe probability is 94.0%, just under the 95% bar I use, so even the training result is borderline once you account for testing 15 configurations. Out of sample the defense failed: in 2022 stocks and long Treasuries fell together, so the TLT/GLD fallback did not protect, and the strategy's drawdown (-25.7%) was worse than SPY's (-20.2%). The honest conclusion: the strategy controlled drawdowns in 2008 but not in 2022, it is not an alpha source, and its edge over SPY is statistically indistinguishable from zero.

Context: Project Hero One's live edge verdict

I hold my own trading system to the same standard. Hero One's forward record from June 30 to August 6, 2026 (paper trading, 35 sessions) was:

Forward record	Value
Scored directional decisions	349
Hit rate	47.9% (148 W / 161 L / 40 breakeven)
Mean return per decision	-0.50%
Session-clustered t-stat	-0.79
Brier score: raw confidence vs constant base rate	0.329 vs 0.244 (confidence carried no information)

The audit also found three estimator defects that were manufacturing false positives. The deflated-Sharpe gate was set at 0.50, the median of the null distribution, so about half of pure-noise strategies passed; I raised it to 0.95, after which 0/36 noise strategies and 12/12 genuine edges were adopted. The isotonic calibration returned the max of tied runs instead of minimizing squared error. A simulator checked stops on closes only and missed 26.7% of real stop events. The verdict: **no edge demonstrated**, and live trading stays off.

Skills demonstrated

Backtesting without look-ahead; train/test discipline; transaction-cost stress; Sharpe, Sortino, Calmar, drawdown; OLS with Newey-West HAC errors; probabilistic and deflated Sharpe ratios (Bailey and Lopez de Prado); block bootstrap; Spearman rank correlation; calibration (Brier score, isotonic regression). The CAPM beta regression for Amphenol (5-year monthly, raw 1.26, R^2 0.50, Blume-adjusted 1.17) is in the APH model's Beta sheet.
Sources: Yahoo Finance adjusted closes; FRED DTB3; Project Hero One reports (strategy_tournament_latest.md, EDGE_VERDICT.md, Aug 7, 2026). Rerun: python quant_study.py; python build_quant_report.py. The last month in the sample (October 2026) is partial, through October 16.